

NEXUS — ENGINEERING COMPLETION & AUDIT REPORT

System Verification, Multi-Tenant RBAC, C++ Engine, Commerce & Quality Gates

EXECUTIVE OVERVIEW & STATUS METRICS

This report provides a full accounting of all engineering implementations, bug fixes, multi-tenant isolation, role-based security, native C++ optimization, and automated quality gates across NEXUS.

- **Active Git Remote:** <https://github.com/aaddi1/NEXUS.git> (Branch: main)
- **Full-Stack Microservices:** REST Gateway (:5000) + Python ML Engine (:8000) + Native C++ Engine + PostgreSQL (:5432) + PWA App (:8001)
- **Quality Gate Matrix:** 15 / 15 Automated Integration Tests Passing (100% Pass Rate via npm test)
- **Super Admin Authority:** Aryan Sharma (aryan@nexus.com / admin123@nexus.com)

1. RESOLVED ERRORS & SYSTEM BUGS INVENTORY

1.1 Frontend & Script Runtime Crashes Resolved

- **SyntaxError in app.js Fixed:** Removed stray object literal snippet left over around line 1850 in the stock transfer logic that was halting browser script execution.
- **ReferenceError (NX_PHOTOS) Fixed:** Corrected unhandled reference to renamed photo catalog on line 2520 with unified alias fallbacks.
- **Dev Overlay Blocker (#nexus-dev-popup) Removed:** Permanently eliminated z-index 999999 development notice dialog that intercepted user mouse clicks.
- **Prefilled Credentials Removed:** Removed hardcoded DEMO_EMAIL and DEMO_PASS inputs. Form now starts empty and clears completely upon logout.
- **Unchecked DOM Manipulations Guarded:** Added safe element verification across all dashboard and table initializers to prevent uncaught TypeError exceptions on startup.

1.2 Security & Authentication Vulnerabilities Remediated

- **Server-Side Role-Based Access Control (RBAC):** Built requireRole, requireSuperAdmin, and requireFounderOrAdmin middlewares. Protected /api/admin behind superadmin checks (returning HTTP 403 Forbidden to unauthorized accounts).
- **Registration Role Escalation Blocked:** Public signup endpoint /api/auth/register strips and normalizes unauthorized role=superadmin attempts to standard Owner defaults.
- **OAuth Identity-Link Table (auth_identities):** Mapped external Google, GitHub, and Microsoft provider accounts to PostgreSQL user records with 30-day JWT sessions.
- **Cryptographic Password Reset Flow:** Built /api/auth/forgot-password and /api/auth/reset-password endpoints utilizing 32-byte hex tokens with 1-hour expiration and bcrypt hashing.
- **Brute-Force Attack Mitigation:** Added sliding-window rate limiting middleware across authentication routes.

NEXUS — COMMERCE, DATABASE & MULTI-TENANT ARCHITECTURE

Row-Level Mutexes, Movement Ledgers, COGS, and Role-Based Portals

2. COMMERCE LOGIC & DATABASE INTEGRITY HARDENING

2.1 Stock Overselling Prevention via Row-Level Mutexes

Order placement previously silently clamped negative stock using `GREATEST(0, quantity - requested)`. The transaction engine has been rebuilt with strict ACID concurrency protection:

- **PostgreSQL Row Lock:** Every product SKU requested locks the exact warehouse stock row via `SELECT quantity FROM inventory WHERE product_id = $1 AND warehouse = $2 FOR UPDATE`.
- **Strict Rejection:** If available stock is less than the requested amount, the transaction rolls back immediately and returns HTTP 400 Insufficient Stock with detailed quantity diagnostics.

2.2 Permanent Inventory Movements Audit Ledger (`inventory_movements`)

Created a persistent movement ledger table tracking every single inventory increment or decrement across all warehouses:

- **Ledger Fields:** `product_id`, `warehouse`, `quantity_change`, `movement_type` (sale, transfer_in, transfer_out, adjustment, restock), `reference_type`, `reference_id`, `actor_id`, and `created_at`.
- **PostgreSQL Parameter Typing Fix:** Resolved critical SQL typing issue by passing signed integer parameters directly via JS parameter bindings rather than inline operator tokens.

2.3 Financial Intelligence, COGS & Payments Synchronization

- **Cost of Goods Sold (COGS):** Added `unit_cost` to products and `order_items` tables so true gross profit and net profit margins can be computed against operating expenses and salaries.
- **Automatic Payment Ledger Reconciliation:** Whenever an order or invoice is settled as paid, a corresponding payment entry is atomically created/updated in the payments table.

3. MULTI-TENANT ISOLATION & ROLE-BASED WORKSPACES

- **Multi-Tenant Data Model:** Added organizations table and `organization_id` foreign keys across all business entities (users, customers, products, categories, inventory, orders, invoices, deals, complaints).
- **Founder / Super Admin Portal:** Executive Dashboard, Multi-Warehouse Stock Transfers, Workforce Management, Salary Disbursements, Operating Expenses, CRM Deals Pipeline, Analytics, and Super Tech Settings.
- **Employee Personal Workspace:** Scoped strictly to personal assignments: My Sales (attributed orders), My Items (issued demonstration stock), My Salary (payment history & salary invoices), My Issues (support tickets), and My Profile.

NEXUS — NATIVE C++ ENGINE, PWA & TEST SUITE

Clang/CMake Compilation, Automated Quality Gates & Seeded Credentials

4. NATIVE C++ HIGH-PERFORMANCE COMPUTATIONAL ENGINE

Built and compiled a native C++ backend computational engine in cpp/ (built with CMake and Clang++ 17):

- **Pricing Engine (cpp/src/pricing_engine.cpp):** Multi-tiered GST tax computation (CGST, SGST, IGST), dual-mode discounts (Fixed INR vs. Percentage), and live currency conversion.
- **Inventory Optimizer (cpp/src/inventory_optimizer.cpp):** Economic Order Quantity (EOQ) formula, safety stock buffer estimation, and multi-depot stock transfer validation.
- **Crypto Engine (cpp/src/crypto_engine.cpp):** Constant-time HMAC-SHA256 vector invoice checksum verification.
- **Executable Benchmark (cpp/src/main.cpp):** Standalone binary nexus_cpp_engine validating all algorithmic mathematical models.

5. AUTOMATED TEST SUITE & QUALITY GATE VERIFICATION

The integration test suite in backend/tests/api.test.js (invoked via npm test) executes 15 critical verification gates:

- ' Backend Health Check (/api/health) — PASS
- ' Super Admin Authentication (aryan@nexus.com) — PASS
- ' Employee Authentication (riya@nexus.com) — PASS
- ' Registration Role Escalation Prevention (role=superadmin blocked) — PASS
- ' RBAC: Employee cannot access /api/admin/stats (HTTP 403) — PASS
- ' RBAC: Super Admin can access /api/admin/stats (HTTP 200) — PASS
- ' Password Reset Request (/api/auth/forgot-password) — PASS
- ' Commerce: Create Order with Atomic Stock Decrement — PASS
- ' Commerce: Reject Order when Stock is Insufficient (HTTP 400) — PASS
- ' Employee Workspace: Scoped Personal Stats & Sales (/api/employees/me/workspace) — PASS
- ' Financial Intelligence Overview (/api/financials/overview) — PASS
- ' Complaints System: Submit Ticket -> Super Admin Resolve — PASS
- ' SSO: OAuth Identity Linking & Audit Log (/api/auth/sso) — PASS
- ' Inventory: Atomic Stock Transfer with ACID Locking (/api/inventory/transfer) — PASS
- ' Invoice: Vector PDF Stream & Cryptographic Verification — PASS

6. DEFAULT SEEDED CREDENTIALS REFERENCE

- **Super Admin (Aryan Sharma):** aryan@nexus.com / aryan123 (or admin123@nexus.com / admin123)
- **Enterprise Company Owner:** reliance@nexus.com / reliance123
- **Retail Shop Owner:** shopowner@nexus.com / shopowner123
- **Sales Lead (Employee):** riya@nexus.com / riya123
- **Inventory Manager (Employee):** rahul@nexus.com / rahul123
- **Finance Lead (Employee):** arjun@nexus.com / arjun123
- **Sales Executive (Employee):** priya@nexus.com / priya123